

HDM propagate the imposed offset until a physical charge anchor is reached because neither has continuous output feedback. HECM uses its voltage residual to correct the explicit SOC state and can therefore re-anchor persistently. DD does not propagate SOC as one isolated state. Its multichannel rolling input allows the prediction to reduce dependence on the disturbed cumulative-charge feature. The changing interaction between that feature and the recurrent context also explains why an initial entry into the recovery band can be followed by relapse. Recovery time, relapse behavior, censoring, and excess-error area together identify HECM as the strongest recovery implementation under this protocol.

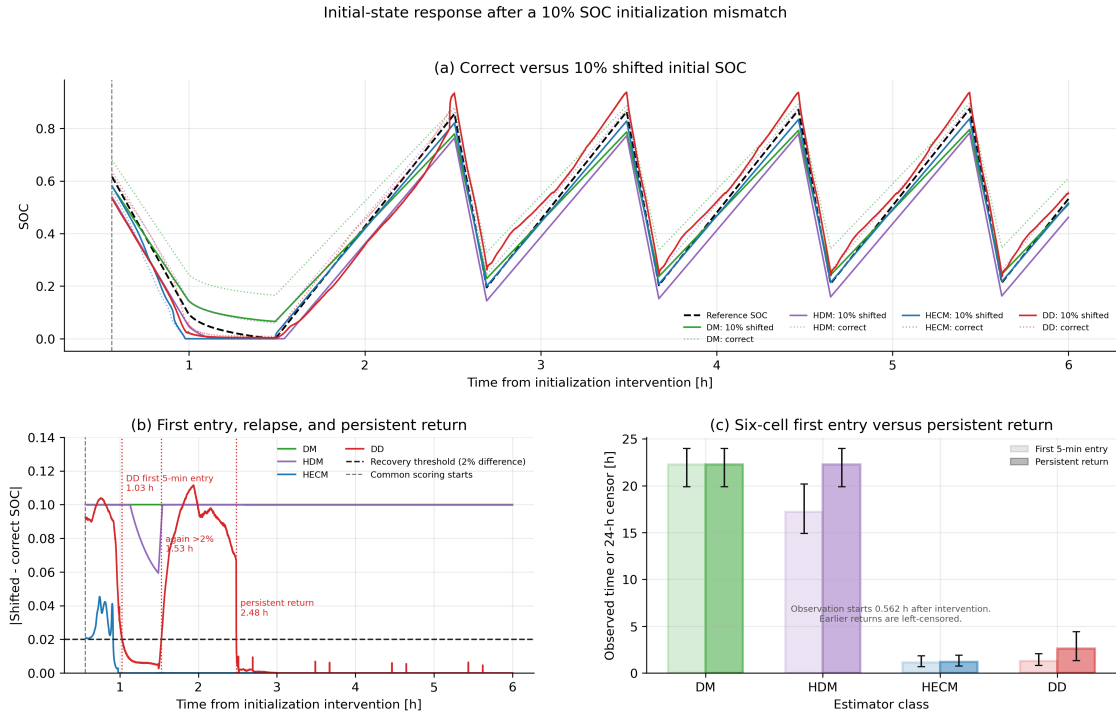


Figure 6.9: Response to a ten-percentage-point SOC-equivalent initialization mismatch. Panels (a) and (b) show correctly and incorrectly initialized trajectories and their absolute paired differences for a representative mid-life window. Panel (c) reports observed first qualifying entry and persistent recovery-or-censor time across the holdout test set with hierarchical 95% confidence intervals.

6.4.6 Missing samples and signal integrity

Figure 6.10 compares periodic and random sample loss with timing jitter. Losing measurement information is substantially more consequential than changing its sampling time. Periodic freezing of every 50th sample increases MAE by 0.0072,